

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of B. Sc. (Biological Sciences) Examination, April 2019

Course code & Name: BS 210 & Analytical Techniques in Biology

Date: 27/04/2019 Day: Saturday Time: 10:00 A.M. To 10:30 A.M. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		20
1.	In SDS-Polyacrylamide Gel Electrophoresis (PAGE), the following reagent is used to reduce disulphide bridges that hold the protein tertiary structure together.		01
	(i) Ammonium persulfate	(ii) β -mercaptoethanol	
	(iii) Sodium dodecyl sulfate	(iv) Ethanol	
2.	The _____ chromatography does not rely on physical properties of the analytes, instead it is based on the unique property of extremely specific biological interactions to achieve analyte separation.		01
	(i) Gel-filtration	(ii) Affinity	
	(iii) Ion-exchange	(iv) Partition	
3.	In size-exclusion chromatography, the _____ will enter into the pores of the gel beads and will therefore migrate at slower rate and elute out last in the separation.		01
	(i) Small analytes	(ii) Large analytes	
	(iii) Intermediate size analytes	(iv) All of them	
4.	In which of the following chromatographic separation, a 30 m long capillary column is housed in an oven which is maintained at 50-300 °C to facilitate analyte volatilization.		01

	(i) HPLC	(ii) Affinity	
	(iii) GC	(iv) Size-exclusion	
5.	A total DNA preparation from <i>E. coli</i> cells was measured using a UV-Visible spectrophotometer and the purity ratio (A_{260}/A_{280}) was determined to be 0.8, this absorbance ratio indicates that the DNA preparation is contaminated with _____.		01
	(i) Proteins	(ii) RNA	
	(iii) Chloroform	(iv) All of them	
6.	In order to break the cells in bacterial genomic DNA isolation, which of the following enzyme is used to digest the peptidoglycan layer in the cell walls.		01
	(i) Cellulase	(ii) Chitinase	
	(iii) Proteinase K	(iv) Lysozyme	
7.	In agarose gel electrophoresis, the following DNA intercalating dye is added to the agarose gels in order to visualize the DNA bands on a UV-trans illuminator.		01
	(i) Bromophenol blue	(ii) Blue dextran	
	(iii) Commasie brilliant blue	(iv) Ethidium bromide	
8.	In density gradient centrifugation biological particles reach _____ position, where the sedimentation rate is zero because their density equals the density of surrounding medium.		01
	(i) Isoelectric	(ii) Isocratic	
	(iii) Isotopic	(iv) Isopycnic	
9.	The following chromatography separates biomolecules based on their net-surface charge.		01
	(i) Affinity	(ii) Gel-filtration	
	(iii) Ion-exchange	(iv) All of them	
10.	The following is a component of the reaction mixture used to perform polymerase chain reaction (PCR).		01
	(i) Taq DNA polymerase	(ii) Oligonucleotide primers	
	(iii) dNTPs	(iv) All of them	
11.	The following salt is most commonly used for concentration of proteins by precipitation.		01
	(i) Magnesium chloride	(ii) Potassium acetate	
	(iii) Sodium chloride	(iv) Ammonium sulphate	
12.	The following method is used for trans-blotting proteins separated by SDS-PAGE to a nitrocellulose membrane for their identification using antibodies.		01
	(i) Eastern blotting	(ii) Western blotting	
	(iii) Southern blotting	(iv) Northern blotting	
13.	Which of the following types of radiation will cause the bonds of a molecule to vibrate?		01
	(i) Fluorescence	(ii) Ultraviolet	
	(iii) Infrared	(iv) Microwave	

14.	Which type of the radiation has highest energy?		01
	(i) IR	(ii) Microwave	
	(iii) UV	(iv) X-ray	
15.	Resolving power of a microscope is a function of –		01
	(i) Wavelength of light used	(ii) Numerical aperture of lens system	
	(iii) Refractive index	(iv) Both (i) and (ii)	
16.	Which molecular electron transition requires the most energy?		01
	(i) $\sigma \rightarrow \sigma^*$	(ii) $\pi \rightarrow \pi^*$	
	(iii) $n \rightarrow \pi^*$	(iv) $n \rightarrow \sigma$	
17.	In microscopy, the maximum numerical aperture (NA) obtainable for an objective is –		01
	(i) 1.2	(ii) 1.4	
	(iii) 1	(iv) 0.6	
18.	Which microscope will be suitable to visualize ribosomal structure inside a cell?		01
	(i) Scanning electron microscope	(ii) Transmission electron microscope	
	(iii) Phase contrast microscope	(iv) Fluorescent microscope	
19.	Which microscope will be suitable for studying protein translocation, localization and interaction?		01
	(i) Dark field microscope	(ii) Fluorescent microscope	
	(iii) Phase contrast microscope	(iv) Scanning electron microscope	
20.	Which of the following spectroscopy does not require a light source (or photons)-		01
	(i) UV-Vis spectroscopy	(ii) Atomic absorption spectroscopy	
	(iii) Flame photometry	(iv) IR spectroscopy	

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Course code & Name: BS 210 & Analytical Techniques in Biology

Date: 27/04/2019 Day: Saturday Time: 10:30 A.M. To 1:00 P.M. Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I		
Q – II Answer the following questions as directed (answer any four).		20
1.	Explain the step-wise method of bacterial genomic DNA purification.	05
2.	Explain the various components of an isocratic HPLC system.	05
3.	Explain the Southern hybridization method.	05
4.	Explain the principle and key steps in gel-permeation chromatography.	05
5.	Explain the alkaline denaturation method for plasmid DNA isolation.	05
6.	What is fluorescence? Explain fluorescence microscopy.	05
SECTION - II		
Q-III Answer the following questions as directed (answer any six).		30
1.	Describe different types of rotors used in biological centrifugation.	05
2.	Explain the principle and key steps of ion-exchange chromatography.	05
3.	Explain the basic instrumentation of electron microscopy.	05
4.	Describe the components and cycling steps of a polymerase chain reaction (PCR).	05
5.	Describe atomic absorption spectroscopy and its application.	05
6.	Write a short note on agarose gel electrophoresis.	05
7.	Describe the ascending, descending, radial and 2D-paper chromatographic techniques.	05
8.	Explain the following: (a) Beer Lambert's law (b) Fluorophores	05